STAT 709 Mathematical Statistics Lecture 22 - Fall 2026

Lecture 22: Confidence sets and asymptotic tests

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Report an issue

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Source and provenance. Most of the contents of the Part VII lectures are taken from Bickel and Doksum (BD), *Mathematical Statistics: Basic Ideas and Selected Topics*, Volume I, second edition (2001) [1], and adapted to the course notation. These lectures are newly prepared; they are **not the original lecture notes from previous years**. Original illustrations, numerical adaptations, and practice problems are identified below.

Lecture guide. This is the second lecture of Part VII and the final lecture of STAT 709. Lecture 21 developed tests and p-values; we now report sets of plausible parameter values, and then use asymptotic sampling laws when exact ones are unavailable. The priority labels follow the course reading conventions.

Reading. BD Section 4.4, pp. 233–241, and Section 4.5, pp. 241–248, supply confidence sets, pivots, and test inversion. For large-sample inference, see Sections 5.4.2–5.4.5, pp. 327–337; Section 6.3, pp. 392–400; and Section 6.4.1, pp. 401–403. Textbook numbers below refer to the 2001 second edition. The Berkeley Stat 210A reader provides a complementary explanation of confidence regions and test inversion [2].

High priority *Overview*

A test addresses a specified hypothesis about the parameter. A confidence set reports the values that remain compatible with the data under a chosen procedure. Both constructions need calibration: a statement about how the procedure behaves under repeated sampling. We first obtain exact coverage from Normal sampling laws and pivots. Then the CLT, Slutsky's theorem, and likelihood expansions give approximate coverage and approximate test levels.

Recap from earlier lectures

From Lectures 10, 11, 13, and 21: the tools used here

Reminder only: this block introduces no new exam requirements.

- **Level and error control**. A level-α test has rejection probability at most α at every null parameter. Used in test inversion.
- **The CLT** and **Slutsky's theorem**. A Normal limit can survive replacement of an unknown scale by a consistent estimate. Used to construct approximate intervals.
- **Fisher information**. Information measures the local sensitivity of the model to its parameter. Used to scale the MLE and compare efficiency.

Learning goals

- Explain what is random in a confidence interval and what its coverage guarantees.
- Construct exact Normal intervals from pivots and invert a family of tests.
- Use a Normal limit and a consistent standard error for approximate inference.
- Explain the regular-model efficiency benchmark and distinguish Wald, score, likelihood-ratio, and Pearson chi-square statistics.

1 Confidence sets and coverage

High priority *What a confidence level means*

Let $X \sim P_\theta$, $\theta \in \Theta$, and let $\tau(\theta)$ be the quantity of interest. The parameter θ specifies the entire data law; $\tau(\theta)$ may be only one component, such as a mean when the variance is also unknown.

Definition 1 (Confidence set; BD Definition 4.4.1 and Section 4.4)**.** *A data-dependent set $C(X)$ is a* ***level-$(1-\alpha)$ confidence set*** *for $\tau(\theta)$ if the relevant events are measurable and*

$$P_\theta\{\tau(\theta) \in C(X)\} \geq 1-\alpha \qquad \text{for every } \theta \in \Theta.$$

For a real-valued target, a set $C(X) = [L(X), U(X)]$ is a ***confidence interval****. A lower confidence bound $L(X)$ satisfies $P_\theta\{L(X) \leq \tau(\theta)\} \geq 1-\alpha$ for every θ; an upper confidence bound is defined with the reverse inequality.*

The **coverage probability** at θ is $P_\theta\{\tau(\theta) \in C(X)\}$. BD calls its infimum over Θ the **confidence coefficient**. For a procedure satisfying this definition, $1-\alpha$ is a guaranteed lower bound, and actual coverage can be larger. A nominal level attached to an approximate interval need not be a finite-sample guarantee; Section 4 states the corresponding limiting claim.

The probability statement concerns $C(X)$ before observing the sample. After observing $X = x$, the reported set $C(x)$ is fixed. Frequentist coverage does not by itself assign probability $1-\alpha$ to the fixed parameter lying in that particular reported set. Such a posterior probability would require a different, Bayesian probability model.

Instructor's Note *Keep the parameter fixed*

Imagine marking the true mean on a number line, then collecting a fresh sample and drawing a new interval each time. The mark stays still; the interval moves. Coverage describes how often this interval-making rule catches the mark. It does not say that every batch of twenty intervals must contain exactly nineteen successes.

1.1 An exact Normal interval

High priority *Known variance*

Suppose $X_1, \ldots, X_n$ are i.i.d. $N(\mu, \sigma^2)$, with known $\sigma > 0$. Write $z_q = \Phi^{-1}(q)$, where Φ is the standard Normal CDF. The sampling law gives

$$Z(\mu) = \frac{\sqrt{n}(\overline{X} - \mu)}{\sigma} \sim N(0,1), \qquad P_\mu\{-z_{1-\alpha/2} \leq Z(\mu) \leq z_{1-\alpha/2}\} = 1-\alpha.$$

Solve the inequality inside the probability for μ. We obtain

$$C(X) = \left[\overline{X} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}},\ \overline{X} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right]. \tag{1}$$

This interval has coverage exactly $1-\alpha$ for every μ; see BD Section 4.4, pp. 234–235. Its half-width is $z_{1-\alpha/2}\sigma/\sqrt{n}$: greater confidence widens the interval, whereas more observations reduce its width.

Continuation of Lecture 21's Normal example. With $n = 25$, $\sigma = 2$, and observed $\overline{x} = .8$, the 95% interval is

$$.8 \pm 1.960\frac{2}{5} \simeq [.016, 1.584].$$

The value 0 is outside it, agreeing with Lecture 21's two-sided test, whose p-value is approximately .0455. This agreement will become a precise test–confidence-set duality in Section 3.

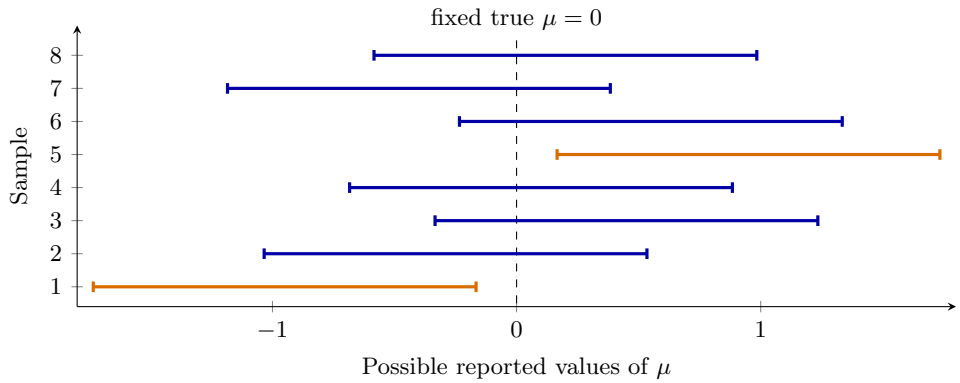


Figure 1: The parameter is fixed; the interval changes with the sample. Each interval uses the same rule $\overline{x} \pm .784$ from the known-variance Normal example. Blue intervals cover 0; orange intervals miss it. These eight selected possible outputs illustrate both outcomes; their fraction of successes is not an estimate of 95% coverage. Original course drawing of the repeated-sampling interpretation in BD Section 4.4.

Exercise prompt. For i.i.d. Normal observations with known $\sigma = 2$, how large must n be so that the 95% interval has half-width at most .25? See Exercise 1 in Section 6 for the full statement and complete solution.

Further reading *Watch repeated intervals*

Kristoffer Magnusson's Interpreting Confidence Intervals repeatedly draws intervals around a fixed population mean and tracks coverage. Compare that repeated-sampling display with the eight selected outputs in Figure 1. Credit: R Psychologist (CC BY).

2 Pivotal quantities

High priority *A distribution that no longer depends on the parameter*

Definition 2 (Pivot; BD Section 4.4, p. 235)**.** *A* ***pivotal quantity*** *is a function $Q(X, \theta)$ whose sampling distribution is known and does not depend on the unknown parameter θ. It may involve unknown parameters, so a pivot need not be a statistic computable from the data alone.*

If $P_\theta\{a \le Q(X, \theta) \le b\} = 1 - \alpha$ for every θ, then

$$C(X) = \{\theta \in \Theta : a \le Q(X, \theta) \le b\}$$

has exact coverage $1 - \alpha$ for the full parameter. When the pivot depends on θ only through a target $\tau(\theta)$, invert it directly for that target. Inversion may produce an interval, a one-sided set, or a more general set; an interval is not automatic. The Normal $Z(\mu)$ above is a pivot.

2.1 The mean when the variance is unknown

High priority *The Student t interval*

Continue with i.i.d. $N(\mu, \sigma^2)$ observations, now with both parameters unknown and $n \ge 2$. Define

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \overline{X})^2.$$

Here the denominator is $n-1$; S^2 is not the variance MLE with denominator n. The Normal sampling identities are

$$Z = \frac{\sqrt{n}(\overline{X}-\mu)}{\sigma} \sim N(0,1), \qquad V = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}, \qquad Z \text{ independent of } V.$$

One way to see the independence is to apply an orthogonal transformation to $((X_1-\mu)/\sigma, \ldots, (X_n-\mu)/\sigma)$, with first row $n^{-1/2}(1,\ldots,1)$. Its coordinates remain independent standard Normals. The first is Z; the sum of squares of the other $n-1$ is V.

By the definition of a Student distribution,

$$T(\mu) = \frac{Z}{\sqrt{V/(n-1)}} = \frac{\sqrt{n}(\overline{X}-\mu)}{S} \sim t_{n-1}.$$

This law does not depend on either unknown parameter. With $t_{\nu,q}$ the q-quantile of t_ν, inversion gives

$$\left[\overline{X} - t_{n-1,1-\alpha/2}\frac{S}{\sqrt{n}},\ \overline{X} + t_{n-1,1-\alpha/2}\frac{S}{\sqrt{n}}\right]. \tag{2}$$

This is BD Example 4.4.1, pp. 235–236. It has exact coverage $1-\alpha$ for every (μ, σ^2) in the Normal model. For $n = 16$, observed $\overline{x} = 5$ and $s = 2$, a 95% interval is $5 \pm 2.131(2/4) \simeq [3.934, 6.066]$ (new numerical values).

Instructor's Note *Estimating the scale costs something*

The denominator S moves from sample to sample too. Student's distribution accounts for that extra variation, using a larger critical value than the Normal distribution at the same confidence level. For large samples the two critical values become close.

The exact t law uses Normal observations. For a general i.i.d. population with finite positive variance, the studentized mean has a standard Normal *limit*; this is an asymptotic justification, not an exact t law.

Mid priority *A pivot for the variance*

BD Example 4.4.2, pp. 236–237, uses the same V to estimate σ^2. Write $\chi^2_{\nu,q}$ for the q-quantile of a chi-square distribution with ν degrees of freedom. Solving

$$\chi^2_{n-1,\alpha/2} \le \frac{(n-1)S^2}{\sigma^2} \le \chi^2_{n-1,1-\alpha/2}$$

gives the exact Normal-model interval

$$\left[\frac{(n-1)S^2}{\chi^2_{n-1,1-\alpha/2}},\ \frac{(n-1)S^2}{\chi^2_{n-1,\alpha/2}}\right]. \tag{3}$$

Notice the reversal of the two quantiles when solving for σ^2. Unlike the mean's studentized Normal limit, this chi-square calibration is not generally rescued by merely assuming finite variance in a non-Normal population.

Exercise prompt. Let $X_1, \ldots, X_n$ be i.i.d. Uniform$(0,\theta)$, $\theta > 0$, and $M = \max_i X_i$. Find the distribution of M/θ and use it to construct an exact $(1-\alpha)$ interval with lower endpoint M. See Exercise 2 and its complete solution.

3 The duality between tests and confidence sets

High priority *Collect the values that are not rejected*

For every proposed target value t, consider a nonrandomized test $\varphi_t(X) \in \{0, 1\}$ of $H_{0,t} : \tau(\theta) = t$, with 1 meaning rejection. The null may be composite because other components of θ are unspecified. Assume that every test has level α over its null:

$$\sup_{\theta:\tau(\theta)=t} P_\theta\{\varphi_t(X) = 1\} \le \alpha.$$

Theorem 1 (Test–confidence-set duality; BD's Duality Theorem, p. 243)**.** *The set $C(X) = \{t : \varphi_t(X) = 0\}$ is a level-$(1-\alpha)$ confidence set for $\tau(\theta)$. Conversely, a level-$(1-\alpha)$ confidence set produces level-α tests by rejecting t exactly when $t \notin C(X)$.*

Proof. For any true parameter θ, the two events $\{\tau(\theta) \notin C(X)\}$ and $\{\varphi_{\tau(\theta)}(X) = 1\}$ are identical. Hence

$$P_\theta\{\tau(\theta) \in C(X)\} = 1 - P_\theta\{\varphi_{\tau(\theta)}(X) = 1\} \ge 1 - \alpha.$$

Read the same identity in reverse for the converse. □

We are inverting a *family* of tests, one for every proposed value, while keeping the observed data fixed. We need not separately correct for multiple testing to obtain this coverage statement: at a fixed true θ, noncoverage is exactly rejection of the one true target value. This is the duality argument, not a claim about simultaneous truth of all possible conclusions. See also the complementary explanation in [2].

3.1 Normal tests, intervals, and p-values

High priority *The same inequality viewed two ways*

In the known-variance Normal model, reject $H_0 : \mu = \mu_0$ against a two-sided alternative when

$$\left|\frac{\sqrt{n}(\overline{X} - \mu_0)}{\sigma}\right| > z_{1-\alpha/2}.$$

For fixed data, the values μ_0 not rejected satisfy $|\overline{x} - \mu_0| \le z_{1-\alpha/2}\sigma/\sqrt{n}$, exactly the interval in (1). The same argument with $T(\mu_0)$ and its t critical value gives (2); this is BD Example 4.5.1, pp. 241–242, including the unknown-variance case.

For the two-sided Normal test, define

$$p_{\mu_0}(x) = 2\left[1 - \Phi\left(\left|\frac{\sqrt{n}(\overline{x} - \mu_0)}{\sigma}\right|\right)\right].$$

With the strict critical-value rejection rule just used, $C(x) = \{\mu_0 : p_{\mu_0}(x) \ge \alpha\}$. If instead rejection is defined by $p_{\mu_0}(x) \le \alpha$, as in Lecture 21's general valid-p-value rule, then inversion gives $\{\mu_0 : p_{\mu_0}(x) > \alpha\}$. Lecture 21 used rejection at $|Z| \ge z_{1-\alpha/2}$ and hence an open interval; the strict $>$ rule here gives a closed interval. These two Normal intervals differ only in their endpoints. For each true μ, equality has probability zero, so both have the same coverage. For discrete data, endpoint conventions and unattainable tail probabilities can matter; retain the exact test rule when inverting it.

A confidence interval and a p-value should agree when they come from the *same* model, test family, tail convention, and calibration. A one-sided test and a two-sided interval need not give matching decisions.

Exercise prompt. Invert the tests that reject $H_0 : \mu = \mu_0$ against $\mu > \mu_0$ when $\sqrt{n}(\overline{X} - \mu_0)/\sigma > z_{1-\alpha}$. Is the result an upper or a lower confidence bound? See Exercise 3 and its complete solution.

4 Large-sample intervals and tests

4.1 An approximate pivot

High priority *Normal limits and estimated standard errors*

An exact pivot is often unavailable. Suppose for each fixed θ that

$$\sqrt{n}(\widehat{\theta}_n - \theta) \xrightarrow{d} N(0, v(\theta)), \qquad \widehat{v}_n \xrightarrow{P} v(\theta) > 0.$$

Define the **estimated standard error** by $\widehat{\mathrm{se}}_n = \sqrt{\widehat{v}_n/n}$. On exceptional samples where $\widehat{v}_n \le 0$, define the procedure arbitrarily; the assumptions make the probability of these samples tend to zero. This convention addresses a limit theorem, not the practical quality of an interval on such a sample.

Proposition 1 (Asymptotic Wald coverage; BD Section 5.4.5)**.** *Under these assumptions,*

$$Z_n(\theta) = \frac{\widehat{\theta}_n - \theta}{\widehat{\mathrm{se}}_n} \xrightarrow{d} N(0,1),$$

and the ***Wald interval***

$$C_n(X) = \left[\widehat{\theta}_n - z_{1-\alpha/2}\widehat{\mathrm{se}}_n, \widehat{\theta}_n + z_{1-\alpha/2}\widehat{\mathrm{se}}_n\right] \tag{4}$$

satisfies $P_\theta\{\theta \in C_n(X)\} \longrightarrow 1-\alpha$.

Proof. Slutsky's theorem gives the asserted Normal limit. On samples where $\widehat{v}_n > 0$, noncoverage is the event $|Z_n(\theta)| > z_{1-\alpha/2}$. The two events can differ only on the exceptional samples, whose probability tends to zero. Since the limiting Normal law has no mass at either cutoff, the noncoverage probability tends to α. □

Likewise, reject $H_0 : \theta = \theta_0$ when $|Z_n(\theta_0)| > z_{1-\alpha/2}$, with approximate two-sided p-value $2[1 - \Phi(|Z_n(\theta_0)|)]$. The null rejection probability tends to α; this is not a finite-sample level guarantee.

The coverage assertion is **pointwise**: it holds at each fixed parameter in the stated model. It does not by itself give uniform coverage over parameters that approach a boundary as n grows. If the target is restricted to Θ, intersecting the interval with Θ does not change whether the true parameter is covered.

High priority *A Bernoulli example*

For i.i.d. Bernoulli(p) data, with fixed $0 < p < 1$, let $\widehat{p} = \overline{X}$. The CLT gives asymptotic variance $p(1-p)$ and consistency gives $\widehat{p}(1-\widehat{p}) \xrightarrow{P} p(1-p)$. Thus an approximate interval is

$$\widehat{p} \pm z_{1-\alpha/2}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}.$$

This is BD Example 4.4.3, specifically equation (4.4.7), p. 238. Do not confuse it with BD's earlier interval obtained by inverting the *null-based* variance $p(1-p)$; Exercise 5 derives that different, score-based interval.

If all trials are failures or all are successes, this estimated standard error is zero. The resulting collapsed interval illustrates why the large-sample approximation can be poor near a boundary. Exact binomial test inversion in BD Example 4.5.2, pp. 244–246, supplies a finite-sample alternative; a large-sample formula alone does not settle those cases.

Exercise prompt. There are 58 successes in 100 independent Bernoulli trials. Find the 95% Wald interval and the associated two-sided Wald test of $p = .5$. See Exercise 4 and its complete solution.

Mid priority *Intervals for a transformed parameter*

Suppose the assumptions for the Wald interval hold at θ, and g is continuously differentiable near θ, with $g'(\theta) \neq 0$. The delta method from Lecture 11 gives

$$\sqrt{n}\{g(\widehat{\theta}_n) - g(\theta)\} \xrightarrow{d} N(0, [g'(\theta)]^2 v(\theta)).$$

Consistency and continuity therefore give the approximate interval

$$g(\widehat{\theta}_n) \ \pm \ z_{1-\alpha/2}|g'(\widehat{\theta}_n)|\widehat{\mathrm{se}}_n$$

for $g(\theta)$. Define the procedure arbitrarily if a finite-sample estimate lies outside the domain of g; at a fixed interior θ, that exceptional probability tends to zero. This extends the asymptotic-pivot construction in BD Section 5.4.5.

For Bernoulli data with $0 < \widehat{p} < 1$, use the log odds $g(p) = \log\{p/(1-p)\}$ and $g'(p) = 1/[p(1-p)]$. The estimated standard error of $g(\widehat{p})$ is $[n\widehat{p}(1-\widehat{p})]^{-1/2}$. For 58 successes in 100 trials, the log-odds interval is $.323 \pm 1.960(.203)$. Transforming its endpoints by $g^{-1}(u) = e^u/(1+e^u)$ gives approximately $[.481, .673]$ for p. A monotone inverse transformation preserves the coverage event, so this is an approximate 95% interval lying inside $(0, 1)$. It need not coincide with the Wald interval computed directly for p; finite-sample Wald intervals depend on the parameterization.

4.2 Likelihood, information, and efficiency

Mid priority *The regular one-parameter likelihood model*

Let $X_1, \ldots, X_n$ be i.i.d. from an identifiable model with density or mass function f_θ, where the scalar parameter lies in an open interval. Write $L_n(\theta) = \prod_{i=1}^n f_\theta(X_i)$ for the sample likelihood and $\ell_n(\theta) = \log L_n(\theta)$ for its logarithm. Thus

$$\ell_n(\theta) = \sum_{i=1}^n \log f_\theta(X_i), \qquad U_n(\theta) = \ell_n'(\theta), \qquad J_n(\theta) = -\frac{1}{n}\ell_n''(\theta).$$

Here U_n is the **score**. Information $I(\theta)$ below is for *one observation*; total sample information is $nI(\theta)$. For the scalar calculations that follow, assume at the true parameter:

1. The true parameter is interior, the model has common support locally, and the log densities are twice continuously differentiable there.
2. The one-observation score $s_\theta(X) = \partial_\theta \log f_\theta(X)$ has mean zero and variance $0 < I(\theta) = E_\theta s_\theta(X)^2 < \infty$. The information identity $I(\theta) = -E_\theta \partial_\theta^2 \log f_\theta(X)$ holds.
3. $J_n(t)$ converges in probability to $I(\theta)$ uniformly as t approaches θ: for every deterministic $\delta_n \downarrow 0$, $\sup_{|t-\theta|\leq\delta_n} |J_n(t) - I(\theta)| \xrightarrow{P} 0$.
4. A consistent MLE $\widehat{\theta}_n$ exists and solves $U_n(\widehat{\theta}_n) = 0$ with probability tending to one.

These are explicit sufficient conditions for the following local argument. BD Sections 5.4.2–5.4.3 give a systematic formulation as conditions A0–A6; they must be checked for the model, not inferred from the name "maximum likelihood."

Mid priority *Asymptotic normality of the MLE*

Under the preceding conditions, the scalar version of BD Theorem 5.4.3, p. 331, gives

$$\sqrt{n}(\widehat{\theta}_n - \theta) \xrightarrow{d} N\left(0, I(\theta)^{-1}\right). \tag{5}$$

Here is the basic argument. Taylor-expand the score about θ:

$$0 = U_n(\widehat{\theta}_n) = U_n(\theta) - nJ_n(\widetilde{\theta}_n)(\widehat{\theta}_n - \theta),$$

with $\widetilde{\theta}_n$ between θ and $\widehat{\theta}_n$. Consistency and the local uniform convergence assumption give $J_n(\widetilde{\theta}_n) \xrightarrow{P} I(\theta)$. The CLT gives $U_n(\theta)/\sqrt{n} \xrightarrow{d} N(0, I(\theta))$; division and Slutsky prove (5). If I is continuous at θ, an estimated standard error is $[nI(\widehat{\theta}_n)]^{-1/2}$. One may instead use $[nJ_n(\widehat{\theta}_n)]^{-1/2}$ when it is positive; local convergence makes this positive with probability tending to one.

What efficiency means here. BD Theorem 5.4.3 also proves an asymptotic variance bound $v(\theta) \geq I(\theta)^{-1}$ for the minimum-contrast estimators satisfying its regularity conditions. The regular MLE achieves that bound; this is **asymptotic efficiency** in that comparison class. It is not a claim of smallest finite-sample mean-squared error, or of optimality among every possible sequence of estimators. BD's Hodges example, p. 332, explains why that qualification matters.

For a fixed alternative $\theta_1 \neq \theta_0$, a consistent estimator with $\widehat{v}_n \xrightarrow{P} v(\theta_1) \in (0, \infty)$ satisfies $|Z_n(\theta_0)| \xrightarrow{P} \infty$. Its two-sided Wald test therefore has power tending to one: **consistency against a fixed alternative**.

Low priority *The information-bound calculation*

The mechanism behind BD's efficiency comparison is Cauchy–Schwarz. Suppose an estimator has expansion

$$\sqrt{n}(\widetilde{\theta}_n - \theta) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} h_\theta(X_i) + o_P(1),$$

where $E_\theta h_\theta = 0$, $E_\theta h_\theta^2 < \infty$, and $E_\theta[h_\theta s_\theta] = 1$. The last identity is a regularity condition; consistency alone does not supply it. Then

$$1 = (E_\theta[h_\theta s_\theta])^2 \leq E_\theta h_\theta^2 \, E_\theta s_\theta^2 = v(\theta) I(\theta).$$

Equality holds when $h_\theta = I(\theta)^{-1} s_\theta$ almost surely, which is the MLE's first-order expansion. This restates the correlation inequality argument in BD's proof on p. 332.

5 Likelihood and chi-square tests

5.1 Calibrating a likelihood ratio

Mid priority *A squared Normal limit*

For $H_0 : \theta = \theta_0$ in the regular scalar model, keep Lecture 21's ratio convention:

$$\Lambda_n = \frac{L_n(\theta_0)}{L_n(\widehat{\theta}_n)} \leq 1, \qquad W_{\mathrm{LR}} = -2 \log \Lambda_n = 2\{\ell_n(\widehat{\theta}_n) - \ell_n(\theta_0)\} \geq 0.$$

BD notation. BD Section 6.3 uses the reciprocal ratio $\Lambda_{\mathrm{BD}} = 1/\Lambda_n \geq 1$. Our $-2 \log \Lambda_n$ is therefore BD's $2 \log \Lambda_{\mathrm{BD}}$; in both conventions, large log-likelihood differences lead to rejection.

Theorem 2 (Scalar Wilks limit; BD Theorem 6.3.1, p. 394)**.** *If the regularity conditions in Section 4.2 hold at θ_0 and H_0 is true, then*

$$W_{\mathrm{LR}} \xrightarrow{d} \chi^2_1.$$

Thus rejecting when $W_{\mathrm{LR}} > \chi^2_{1,1-\alpha}$ has limiting null rejection probability α; its approximate p-value is $P\{\chi^2_1 \geq W_{\mathrm{LR,obs}}\}$.

The explanation is local quadratic curvature: twice the log-likelihood loss is approximately the squared estimation error measured in information units, $nI(\theta_0)(\widehat{\theta}_n - \theta_0)^2$. The MLE limit makes that quantity a squared standard Normal, hence a χ^2_1 limit.

In Lecture 21's known-variance Normal mean example, $W_{\mathrm{LR}} = n(\overline{X} - \mu_0)^2/\sigma^2 = Z^2$ has a χ^2_1 law *exactly*. Wilks' result is the large-sample extension; exactness should not be carried over to a general likelihood model.

Low priority *The scalar likelihood-ratio proof*

Taylor-expand $\ell_n(\theta_0)$ about its interior maximizer $\widehat{\theta}_n$. The first derivative there is zero, so for some θ^*_n between them,

$$W_{\mathrm{LR}} = nJ_n(\theta^*_n)(\widehat{\theta}_n - \theta_0)^2.$$

Under H_0, consistency gives $\theta^*_n \xrightarrow{P} \theta_0$, and local uniform convergence gives $J_n(\theta^*_n) \xrightarrow{P} I(\theta_0)$. Equation (5), the continuous mapping theorem, and Slutsky now give $W_{\mathrm{LR}} \xrightarrow{d} Z^2$ for $Z \sim N(0,1)$. This is the one-dimensional version of BD's proof on p. 394.

5.2 Wald, score, and likelihood-ratio statistics

Mid priority *Three ways to compare a null with the data*

Assume the same regular scalar model and continuity of I. For the point null $\theta = \theta_0$, the standard statistics are

Test	Statistic	What is measured?
Wald	$W_{\mathrm{W}} = nI(\widehat{\theta}_n)(\widehat{\theta}_n - \theta_0)^2$	Distance of estimate from null
Score (Rao)	$W_{\mathrm{S}} = U_n(\theta_0)^2/[nI(\theta_0)]$	Slope at the null
Likelihood ratio	$W_{\mathrm{LR}} = 2\{\ell_n(\widehat{\theta}_n) - \ell_n(\theta_0)\}$	Loss from imposing the null

Each tends to χ^2_1 under H_0, and their pairwise differences tend to zero in probability under H_0. For Wald this follows from (5); for score it follows from the score CLT. The score expansion used for the MLE relates the two, and the quadratic likelihood expansion relates them to LR. See BD Sections 5.4.4 and 6.3.2. Finite-sample values and confidence sets can nevertheless differ.

For Bernoulli data, $I(p) = 1/[p(1-p)]$ and $U_n(p_0) = n(\widehat{p} - p_0)/[p_0(1-p_0)]$. Hence

$$W_{\mathrm{W}} = \frac{n(\widehat{p} - p_0)^2}{\widehat{p}(1-\widehat{p})}, \qquad W_{\mathrm{S}} = \frac{n(\widehat{p} - p_0)^2}{p_0(1-p_0)},$$

$$W_{\mathrm{LR}} = 2n\left\{\widehat{p}\log\frac{\widehat{p}}{p_0} + (1-\widehat{p})\log\frac{1-\widehat{p}}{1-p_0}\right\}.$$

These formulas distinguish the estimated variance from the null variance; all require appropriate calibration. The regular asymptotic argument fixes $0 < p_0 < 1$. Use $0\log 0 = 0$ to evaluate the LR expression at an observed boundary; the Wald statistic is undefined there. Exercise 5 derives the score interval from this formula.

5.3 Likelihood confidence sets

Mid priority *Invert the likelihood-ratio tests*

The same duality applies to approximate tests. For each candidate t in the regular scalar model, write $W_{\mathrm{LR}}(t) = 2\{\ell_n(\widehat{\theta}_n) - \ell_n(t)\}$. The set

$$C_n(X) = \{t : W_{\mathrm{LR}}(t) \le \chi^2_{1,1-\alpha}\}$$

has limiting coverage $1-\alpha$ at each fixed true interior parameter where the Wilks limit holds. Indeed, coverage at θ is exactly the event $W_{\mathrm{LR}}(\theta) \le \chi^2_{1,1-\alpha}$. This combines BD's duality theorem with its likelihood-ratio limit.

The quadratic approximation to the log likelihood gives the corresponding Wald interval locally, since $\chi^2_{1,1-\alpha} = z^2_{1-\alpha/2}$. The actual likelihood confidence set can be asymmetric and need not be an interval. For a scalar target with nuisance parameters, the appropriate extension maximizes over the nuisance parameters at each candidate value; its calibration needs the additional regular multiparameter conditions.

Mid priority *Compare the intervals numerically*

The same data, three methods. For 58 successes in 100 trials, test $p_0 = .5$ and report 95% intervals. The following are original numerical evaluations of the three formulas; all p-values use a χ^2_1 calibration.

Method	Statistic	p-value	95% interval
Wald	2.627	.105	[.483, .677]
Score (Wilson interval)	2.560	.110	[.482, .672]
Likelihood ratio	2.571	.109	[.482, .674]

The Wald calculation is worked out in Exercise 4; Exercise 5 derives the Wilson roots. For the likelihood interval, solve $W_{\mathrm{LR}}(p) = \chi^2_{1,.95} \simeq 3.841$ on each side of $\widehat{p} = .58$. All three intervals contain .5, agreeing with the three nonrejections at level .05.

With only 3 successes in 20 trials, the difference is more visible:

Method	95% interval
Wald (before intersecting with $[0,1]$)	$[-.006, .306]$
Score (Wilson interval)	[.052, .360]
Likelihood ratio	[.040, .344]

Use the same formulas with $n = 20$ and $\widehat{p} = .15$. The likelihood interval is asymmetric, and the unmodified Wald interval extends below zero. Staying inside the parameter space does not by itself guarantee finite-sample coverage: all three use asymptotic calibration here.

Further reading *Several restrictions and the limits of the approximation*

For a fixed-dimensional regular model with d parameters and a smooth interior null imposing r independent restrictions, BD Theorems 6.3.2–6.3.3, pp. 395–397, give $-2\log\Lambda_n \xrightarrow{d} \chi^2_r$ under the null. Here r is the dimension lost under the null, not the number of observations or the number of fitted parameters. The restricted MLE and the null model must satisfy the corresponding regularity conditions.

Boundary nulls, nonidentifiability, and parameter-dependent support can invalidate this law. For example, the Uniform endpoint model of Exercise 2 has parameter-dependent support; its exact pivot does not justify a routine χ^2_1 likelihood-ratio approximation. The multiparameter proof and nonregular limits are further reading.

An exact nonregular likelihood-ratio law. In the Uniform endpoint model, set $M = \max_i X_i$ and test $H_0 : \theta = \theta_0$ against $\theta \neq \theta_0$. The likelihood is proportional to θ^{-n} for $\theta \geq M$. Using suprema, its unrestricted maximum value is M^{-n}, so under H_0

$$\Lambda_n = (M/\theta_0)^n, \qquad W_{\mathrm{LR}} = 2n \log(\theta_0/M).$$

If an observed $M > \theta_0$, the null likelihood is zero and $W_{\mathrm{LR}} = \infty$; this event has null probability zero. Exercise 2 shows that $U = (M/\theta_0)^n \sim \mathrm{Uniform}(0,1)$ under H_0. Consequently, for $w \geq 0$,

$$P_{\theta_0}(W_{\mathrm{LR}} > w) = P(U < e^{-w/2}) = e^{-w/2}.$$

Thus $W_{\mathrm{LR}} \sim \chi^2_2$ exactly for every n, despite the single parameter. A routine χ^2_1 cutoff 3.841 would give rejection probability $e^{-3.841/2} \simeq .1465$, not .05. The support changes with θ, violating the regularity assumptions. This original calculation explains the warning using the lecture's exact pivot.

Further reading *Explore likelihood, Wald, and score*

Kristoffer Magnusson's Understanding Maximum Likelihood Estimation visualizes the three tests and their intervals in a Normal model with unknown mean and variance. When testing the mean, it maximizes over the nuisance variance to form a profile likelihood. This illustrates the nuisance-parameter extension mentioned above. Credit: R Psychologist; visualization CC0, explanatory text CC BY 4.0. The displayed small-sample results still use their stated calibrations; the visualization does not make Wilks' approximation exact.

5.4 Pearson's chi-square goodness-of-fit test

Mid priority *Observed and expected category counts*

Suppose $N = (N_1, \ldots, N_k) \sim \mathrm{Multinomial}(n, p)$ and test $H_0 : p = p^0$, where the k probabilities $p^0_j > 0$ are specified in advance and sum to one. Keep k fixed as $n \to \infty$. The **Pearson chi-square statistic** is

$$X^2 = \sum_{j=1}^{k} \frac{(N_j - np^0_j)^2}{np^0_j} \xrightarrow[H_0]{d} \chi^2_{k-1}. \tag{6}$$

There are $k-1$ degrees of freedom because the k count deviations sum to zero. Reject for $X^2 > \chi^2_{k-1,1-\alpha}$; the approximate p-value is the upper tail of that chi-square law. This is BD Section 6.4.1, pp. 401–403, equation (6.4.1).

For the same multinomial model, the likelihood-ratio statistic is

$$G^2 = 2\sum_{j=1}^{k} N_j \log \frac{N_j}{np^0_j}, \qquad 0 \log 0 := 0.$$

A Taylor expansion about $N_j = np^0_j$, with the linear terms cancelling, gives $G^2 = X^2 + o_P(1)$ under the fixed interior null. Thus both use the same limiting calibration. Small expected counts can make that approximation poor; the fixed-positive-probability assumption makes every np^0_j grow in the stated limit. If model parameters are estimated from these counts, the simple-null $k-1$ rule cannot be reused without accounting for the fitted constraints (BD Section 6.4.2).

BD Example 6.4.1: Mendel's four categories. BD reports the counts below, with theoretical probabilities $(9, 3, 3, 1)/16$ and total $n = 556$.

Category	Observed	Expected under H_0
Round yellow	315	312.75
Wrinkled yellow	101	104.25
Round green	108	104.25
Wrinkled green	32	34.75

The statistic is $X^2 \simeq .470$, with three degrees of freedom and approximate p-value .925. The likelihood-ratio calculation gives $G^2 \simeq .475$ and an approximate p-value of .924, a similar conclusion. Thus neither test rejects at level .05. This is a statement about discrepancy under the model, not a posterior probability that the theory is true. The counts and model are from BD, p. 403; the displayed numerical tail is recalculated here.

Low priority *Why one degree of freedom is lost*

Under the multinomial CLT, the standardized residual vector $Y_{n,j} = (N_j - np_j^0)/\sqrt{np_j^0}$ converges to a centered Normal vector with covariance $I_k - aa^T$, where $a_j = \sqrt{p_j^0}$ and $a^T a = 1$. This matrix is the orthogonal projection onto the $(k-1)$-dimensional space perpendicular to a. In orthonormal coordinates on that space, the limit has $k-1$ independent standard Normal components and one zero component. Therefore $X^2 = Y_n^T Y_n \xrightarrow{d} \chi^2_{k-1}$. This is an elementary CLT/projection justification of the calibration in BD Section 6.4.1.

Key ideas

- Coverage describes a random set under repeated sampling; the parameter is fixed.
- Exact pivots give finite-sample coverage. Approximate pivots require a limit argument and a reliable standard error.
- Inverting level-α tests gives a level-$(1-\alpha)$ confidence set for the tested target.
- Regular likelihood theory connects information, efficient estimation, and Wald, score, and likelihood-ratio tests.
- Chi-square degrees of freedom come from the model's constraints; exact and asymptotic calibrations must remain distinct.

Looking ahead

This completes STAT 709. The course's recurring chain is: specify a model, understand a statistic's sampling law, and use that law to assess estimation error or calibrate a decision. Further study extends these ideas to richer models and more demanding settings; the habit to retain is reporting the estimate, its uncertainty, and the assumptions that justify both. Review the two lectures through the Part VII checkpoint.

Supplemental reference: key notions and definitions

The rows retain the priority of their source blocks. This is a reference guide, not a replacement for the hypotheses of the results.

Notion	Definition or key fact	Tier
Confidence set	$P_\theta\{\tau(\theta) \in C(X)\} \geq 1-\alpha$ for every θ. The set is random.	High
Coverage probability and coefficient	Coverage is the probability at a given parameter; the coefficient is its infimum over the model.	High
One-sided bound	A lower bound satisfies $P_\theta\{L(X) \leq \tau(\theta)\} \geq 1-\alpha$; an upper bound reverses the inequality.	High
Normal mean interval	Known variance: $\overline{X} \pm z_{1-\alpha/2}\sigma/\sqrt{n}$, exact under Normal sampling.	High
Pivotal quantity	$Q(X,\theta)$ has a known parameter-free law; it need not be a statistic.	High
Student pivot and interval	$\sqrt{n}(\overline{X}-\mu)/S \sim t_{n-1}$ exactly for Normal data; use $S^2 = \sum_i (X_i - \overline{X})^2/(n-1)$.	High
Variance pivot	$(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$ in the Normal model.	Mid
Test inversion	$C(X) = \{t : \varphi_t(X) = 0\}$; noncoverage is rejection of the true target value.	High
Intervals and p-values	Invert the same test family and boundary convention. With rejection at $p_t < \alpha$, retain $p_t \geq \alpha$.	High
Estimated standard error	$\widehat{\mathrm{se}} = \sqrt{\widehat{v}_n/n}$ estimates the scale of the estimator's sampling fluctuation.	High
Wald interval and test	$\widehat{\theta}_n \pm z_{1-\alpha/2}\widehat{\mathrm{se}}$ has limiting coverage $1-\alpha$ under the stated CLT and consistency assumptions.	High
Pointwise asymptotic coverage	Coverage tends to $1-\alpha$ for each fixed parameter; this need not be uniform over moving parameters.	High
Transformed-parameter interval	Use $g(\widehat{\theta}) \pm z\lvert g'(\widehat{\theta})\rvert\widehat{\mathrm{se}}$ under the delta-method assumptions; a monotone inverse preserves the coverage event.	Mid
Score and information	$U_n = \ell_n'$; $I(\theta)$ is information per observation, and $nI(\theta)$ is total information.	Mid
Regular MLE limit	$\sqrt{n}(\widehat{\theta}_n - \theta) \Rightarrow N(0, I(\theta)^{-1})$, subject to regularity and consistency.	Mid
Asymptotic efficiency	Achieving the information bound within the stated regular comparison class, not universal finite-sample optimality.	Mid
Information-bound proof	An asymptotic linear expansion and $E(hs) = 1$ give $vI \geq 1$ by Cauchy–Schwarz.	Low
Scalar likelihood-ratio limit	$-2\log\Lambda_n \Rightarrow \chi^2_1$ under a regular scalar point null; BD uses the reciprocal ratio.	Mid
Scalar Wilks proof	Expand the log likelihood at the MLE, then use its Normal limit and local curvature convergence.	Low
Wald and score statistics	$nI(\widehat{\theta}_n)(\widehat{\theta}_n - \theta_0)^2$ and $U_n(\theta_0)^2/[nI(\theta_0)]$ share the regular χ^2_1 limit.	Mid
Likelihood confidence set	Retain candidates whose likelihood-ratio statistics are at most the appropriate chi-square critical value; coverage is asymptotic.	Mid
Pearson goodness of fit	$X^2 = \sum_j (N_j - np_j^0)^2/(np_j^0) \Rightarrow \chi^2_{k-1}$ for a fixed, fully specified, positive multinomial null.	Mid
Degrees-of-freedom argument	The limiting standardized count residuals occupy a $(k-1)$-dimensional subspace.	Low
A nonregular likelihood ratio	For the Uniform endpoint, $-2\log\Lambda_n \sim \chi^2_2$ exactly. Parameter-dependent support invalidates regular dimension counting.	Further
Several restrictions	Under the additional regular multiparameter conditions, r independent restrictions give a χ^2_r limit.	Further

6 Exercises and complete solutions

High priority *Plan precision before collecting data*

Exercise 1 (Sample size for a target half-width). Original practice using the Normal interval. Let X_i be i.i.d. $N(\mu, 4)$. Find the smallest integer n for which the usual 95% interval has half-width at most .25. Explain why the answer does not require knowing μ or observing $\overline{X}$.

Solution. The half-width is $z_{.975}2/\sqrt{n}$. Thus

$$n \geq \left(\frac{2z_{.975}}{.25}\right)^2 \simeq 245.85,$$

so the smallest integer is 246. The half-width depends only on the known scale, chosen confidence level, and sample size. The unknown mean and the observed sample mean locate the interval but do not change its width. □

Mid priority *An exact pivot outside the Normal model*

Exercise 2 (An exact interval for a Uniform endpoint). Original pivotal practice. Suppose $X_1, \ldots, X_n$ are i.i.d. Uniform$(0, \theta)$, $\theta > 0$, and $M = \max_i X_i$. Find the CDF of $Q = M/\theta$. Construct an exact level-$(1-\alpha)$ confidence interval for θ with lower endpoint M, and verify its coverage.

Solution. For $0 \leq u \leq 1$, independence gives

$$P_\theta(M/\theta \leq u) = \prod_{i=1}^{n} P_\theta(X_i \leq \theta u) = u^n.$$

The CDF is zero below zero and one above one, so this is a pivot. Let $a = \alpha^{1/n}$. Its continuous distribution gives $P_\theta(a \leq M/\theta \leq 1) = 1 - a^n = 1 - \alpha$. Solving both inequalities for θ yields

$$C(X) = [M, M/\alpha^{1/n}].$$

Equivalently, $M \leq \theta$ almost surely, and the upper endpoint covers θ exactly when $M/\theta \geq \alpha^{1/n}$, which has probability $1 - \alpha$. The endpoints need not be symmetric about an estimator. □

High priority *Check the direction of inversion*

Exercise 3 (Inverting one-sided tests). Original practice using BD's duality theorem. For i.i.d. $N(\mu, \sigma^2)$ data with known $\sigma > 0$, invert the level-α tests that reject $\mu = \mu_0$ against $\mu > \mu_0$ when $\sqrt{n}(\overline{X} - \mu_0)/\sigma > z_{1-\alpha}$. Give the resulting one-sided confidence set and verify coverage directly. For $n = 25$, $\sigma = 2$, $\overline{x} = .8$, compute the 95% bound.

Solution. Nonrejection is equivalent to

$$\mu_0 \geq \overline{X} - z_{1-\alpha}\frac{\sigma}{\sqrt{n}}.$$

Thus the confidence set is $[L(X), \infty)$ with $L(X) = \overline{X} - z_{1-\alpha}\sigma/\sqrt{n}$, a lower bound. Directly,

$$P_\mu\{L(X) \leq \mu\} = P_\mu\left\{\frac{\sqrt{n}(\overline{X} - \mu)}{\sigma} \leq z_{1-\alpha}\right\} = 1 - \alpha.$$

Numerically $L(x) = .8 - 1.644854(2/5) \simeq .1421$. The one-sided 95% bound uses $z_{.95}$, whereas a two-sided 95% interval uses $z_{.975}$. □

High priority *An approximate interval and its matching test*

Exercise 4 (Wald interval and test for a proportion). Adapted from BD Example 4.4.3; new numerical values. There are 58 successes in 100 independent Bernoulli trials. Compute a 95% Wald interval for p and the two-sided Wald test statistic and approximate p-value for $H_0 : p = .5$. Check that the interval and test agree, and identify the approximation used.

Solution. Here $\widehat{p} = .58$ and $\widehat{\mathrm{se}} = \sqrt{.58(.42)/100} \simeq .04936$. The interval is

$$.58 \pm 1.959964(.04936) \simeq [.4833, .6767].$$

The Wald Normal statistic is $Z = (.58 - .5)/.04936 \simeq 1.621$, so $W_{\mathrm{W}} = Z^2 \simeq 2.6273$ and the approximate two-sided p-value is $2[1 - \Phi(1.621)] \simeq .1050$. The test does not reject at .05, and the interval contains .5. Both calculations use the Bernoulli CLT and a consistent plug-in variance; neither is an exact binomial calculation. □

Mid priority *Inverting a different variance choice*

Exercise 5 (The Wilson score interval). BD Example 4.4.3, equations (4.4.3)–(4.4.4), restated. Let $\widehat{p} = S/n$ in the Bernoulli model and $z = z_{1-\alpha/2}$. Invert the score-test inequalities $n(\widehat{p} - p)^2 \le z^2 p(1 - p)$ for candidate $0 < p < 1$. Give the two roots and the closed interval obtained by adjoining its endpoints. Compare this with the Wald interval, including the case $S = 0$.

Solution. Expand and gather terms:

$$(n + z^2)p^2 - (2n\widehat{p} + z^2)p + n\widehat{p}^2 \le 0.$$

The leading coefficient is positive. The quadratic formula therefore gives the interval between the roots

$$\frac{\widehat{p} + z^2/(2n) \;\pm\; z\sqrt{\widehat{p}(1 - \widehat{p})/n + z^2/(4n^2)}}{1 + z^2/n}.$$

If $0 < \widehat{p} < 1$, the quadratic is positive at 0 and 1 and strictly negative at $\widehat{p}$, so one root lies in $(0, \widehat{p})$ and the other in $(\widehat{p}, 1)$. At $\widehat{p} = 0$, its roots are 0 and $z^2/(n + z^2)$; at $\widehat{p} = 1$, they are $n/(n + z^2)$ and 1. Thus both roots lie in $[0, 1]$ in every case. The interval's intersection with $(0, 1)$ is the inverted set in the stated interior model; taking its closure supplies the displayed endpoints. This is commonly called the **Wilson score interval**. It differs from the Wald interval because candidate p, rather than $\widehat{p}$, is used in the variance while inverting.

When $S = 0$, the score interval is $[0, z^2/(n + z^2)]$, whereas the unmodified Wald formula collapses to $[0, 0]$. This contrast does not make the score interval exact: its nominal coverage still relies on approximate calibration. Exact binomial inversion is a separate construction. □

Mid priority *A simple multinomial null*

Exercise 6 (Pearson testing with four categories). Original practice using BD Section 6.4.1. In 100 independent trials with four possible categories, the observed counts are $(18, 22, 28, 32)$. Test the hypothesis that all four probabilities are $1/4$ using Pearson's statistic. State the degrees of freedom, compare with $\chi^2_{3,.95} \simeq 7.815$, and interpret the conclusion. Also compute G^2 and compare its approximate p-value.

Solution. Every expected count is 25. Therefore

$$X^2 = \frac{(-7)^2 + (-3)^2 + 3^2 + 7^2}{25} = 4.64.$$

The null specifies all category probabilities, and the count total removes one degree of freedom, giving $k-1=3$. Since $4.64 < 7.815$, the test does not reject at approximate level .05. Its approximate p-value is $P(\chi^2_3 \geq 4.64) \simeq .2001$. This means the observed discrepancy is not large by this test's calibration; it does not prove equal probabilities. For the likelihood-ratio comparison,

$$G^2 = 2\{18\log(18/25) + 22\log(22/25) + 28\log(28/25) + 32\log(32/25)\} \simeq 4.695.$$

Its χ^2_3 upper-tail probability is approximately .1956, again giving nonrejection. Both chi-square calibrations are large-sample approximations here. □

References

[1] Peter J. Bickel and Kjell A. Doksum. *Mathematical Statistics: Basic Ideas and Selected Topics, Volume I.* Prentice Hall, Upper Saddle River, NJ, 2 edition, 2001.

[2] UC Berkeley Stat 210A. p-values, confidence regions, and (mis-)interpreting tests. Fall 2025 course reader, "Confidence Regions", 2025. Complementary explanation of coverage and test inversion. `https://stat210a.berkeley.edu/fall-2025/reader/testing-interpretation.html`. Accessed October 27, 2026.

What changed

October 27, 2026: added confidence sets, asymptotic inference, the original coverage illustration, six complete exercises, and a supplemental reference. October 28, 2026: clarified coverage guarantees, exceptional samples in the Wald proof, likelihood notation, and the score interval's boundary cases. October 29, 2026: added a delta-method interval, numerical comparisons of Wald, Wilson, and likelihood intervals, an exact nonregular likelihood-ratio counterexample, and credited visual explorations. Clarified endpoint conventions, reading ranges, and exercise titles.